TARUN VUSIRIKAYALA

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PROFESSIONAL SUMMARY

Data Analyst with experience delivering data-driven solutions through healthcare and financial services. Expertise in clinical analytics, predictive modeling, risk stratification, and executive reporting using SQL, Python, SAS, Power BI, and Tableau. Demonstrated ability to transform large-scale medical claims, patient, and financial datasets into actionable insights that improve program performance, operational efficiency, and regulatory compliance. Adept at collaborating with clinical, finance, and compliance stakeholders to deliver analytical solutions in fast-paced, regulated environments.

TECHNICAL SKILLS

Languages & Tools: SQL, Python, R, SAS, VBA, Pandas, NumPy, Scikit-learn, Statsmodels, TensorFlow, NLTK
Reporting & Dashboards: Power BI, Tableau, Excel (Power Query, Pivot Tables), Visio
Data Management: ETL Pipelines, Data Validation, Reconciliation, Anomaly Detection
Databases & Warehousing: AWS Redshift, Microsoft SQL Server, Snowflake, PostgreSQL, Azure SQL
Cloud & Platforms: AWS, Microsoft Azure, SharePoint, Confluence
Version Control & Delivery: Git, GitHub, Jira, Agile/Scrum, SDLC, REST APIs
Modeling & Analytics: Predictive Modeling, Risk Stratification, Patient Outcome Modeling, EDA, Active Learning, Financial Modeling
Domain Knowledge: Clinical Data, Claims Analytics, Credit Risk, Regulatory Reporting, CMS, HIPAA

PROFESSIONAL EXPERIENCE

Senior Data Analyst, Cigna Healthcare *| March 2025 – Present*

- Partnered with clinical leadership to gather and document business requirements for new disease management and behavioral health programs, leveraging Visio to map workflows and ensure compliance alignment.
- Utilized SQL and SAS for patient outcome modeling and risk stratification, improving identification of high-risk cohorts and increasing program engagement by 15%.
- Developed Power BI dashboards for tracking clinical KPIs (readmission rates, care gaps, medication adherence), driving early interventions that reduced adverse patient events by 12%.
- Applied cost-benefit analysis and Excel-based financial modeling to optimize eligibility rules and resource allocation, enhancing operational efficiency by 18%.
- Leveraged AWS Redshift to manage large-scale healthcare data pipelines, supporting advanced clinical data analytics and reducing reporting latency by 25%.
- Created Power BI visualizations and utilization management reports to present program performance insights to clinical managers, cutting manual reporting time by 30% and improving decision support.
- Supported organizational change readiness by developing training materials and SOPs in SharePoint/Confluence for 150+ care managers, ensuring seamless adoption of new clinical protocols.
- Conducted regulatory gap analysis and compliance monitoring, achieving 100% CMS/HIPAA audit compliance and strengthening clinical program governance.

Graduate Assistant, SUNY INSTITUTE OF TECHNOLOGY | *Sep 2023 - Apr 2024*

- Designed and delivered data analysis coursework focusing on SQL, Python, and statistical modeling, emphasizing practical business applications and reporting techniques
- Mentored graduate students on translating analytical findings into business recommendations and enterprise-level reporting strategies
- Supervised development of REST-based APIs and microservice analytics solutions, emphasizing scalability and operational excellence
- Created educational materials using PowerPoint and Excel to demonstrate advanced analytical techniques and data visualization best practices

Data Analyst, HSBC Global Banking and Markets *| January 2021 – December 2022*

- Analyzed customer transactions, loan portfolios, and risk assessment data across multiple enterprise systems using Python and SQL, improving data accessibility by 45% and enabling near real-time financial reporting across global banking operations.
- Designed and delivered interactive Power BI and Tableau dashboards providing executives with dynamic views of revenue growth, credit exposure, and operational efficiency metrics, accelerating data-driven decision-making across business lines.
- Developed predictive models using Python, Scikit-learn, and Stats models to forecast loan defaults, customer churn, and portfolio performance, improving risk management accuracy by 20% and reducing financial exposure across credit portfolios.
- Partnered with Global Banking, Finance, and Transformation teams to define reporting requirements, standardize KPI frameworks, and ensure all analytical deliverables aligned with internal data governance policies and regulatory standards.
- Implemented automated data quality workflows incorporating validation scripts, anomaly detection, and reconciliation checks, reducing data inconsistencies by 30% and strengthening compliance across enterprise reporting systems.
- Optimized complex SQL queries and reporting workflows within the financial data warehouse, improving query performance by 40% and reducing dashboard load times for over 600 internal business users.
- Conducted monthly trend analysis and performance benchmarking across loan and transaction datasets, producing executive-level reports that directly informed credit risk decisions and operational strategy for senior leadership.

Data Science Intern, Modak *| July 2020 – December 2020*

- Conducted literature reviews on active learning methodologies and recommended integration strategies for the existing machine learning model.
- Investigated innovative labeling approaches for training and evaluation datasets, improving label consistency and overall data quality.
- Built and implemented an end-to-end active learning pipeline, analyzing performance outcomes and documenting model accuracy improvements.
- Explored label quality enhancement strategies to reduce annotation noise and strengthen training data accuracy across machine learning workflows.

PROJECTS

Risk Reporting Automation Platform *(Python, SQL, Power BI, Pandas)*

- Engineered a Python-based automation framework for operational risk data ingestion, anomaly detection, and management report generation, cutting analysis cycle time and standardizing threshold-breach escalation across structured datasets.
- Reduced manual monthly reporting effort by centralizing output formatting used in operational risk governance reviews, replacing multiple spreadsheet-based processes with a single automated pipeline.

Healthcare Data Pattern Analysis *(Python, NLP, Scikit-learn, TensorFlow, NLTK, Pandas)*

- Built NLP and machine learning models to extract structured clinical risk indicators from unstructured healthcare records, producing compliance-ready outputs for downstream quality measurement and regulatory reporting workflows.
- Applied feature engineering and multi-class classification techniques to improve model stability and analytical reliability, supporting both clinical decision-making and compliance data quality initiatives.

EDUCATION

SUNY Institute of Technology *| Master of Science, Computer Science | Jan 2023 – Dec 2024 | Utica, New York*
CVR College of Engineering *| Bachelor of Technology, Computer Science | Aug 2018 – Apr 2022 | Hyderabad, India*